Što su Transformeri?

Transformeri su arhitektura neuronske mreže koja se koristi u području NLP-a i computer visiona.

2017. godine Vaswani et al. su objavili članak "Attention is all you need" u kojem je predstavljena arhitektura tranformera

https://arxiv.org/pdf/1706.03762

Arhitektura transformera koristi "attention" mehanizam kako bi se obradila čitava rečenica odjedanput umjesto da se obrađuje riječ po riječ

Primjer:

XYZ went to France in 2019 and there he met the president of that country

Riječ "that" se odnosi na koju zemlju? Klasične metode koje smo do sada radili ne mogu uhvatiti značenje te riječi.

RNN & LSTM

Nakon klasičnih metoda poput TF-IDF i Word2Vec, sljedeći korak u razvoju NLP-a bili su modeli temeljeni na neuronskim mrežama, poput RNN-a i LSTM-a. Oni su prvi pokušali uvesti redoslijed riječi i kontekst u obradu teksta, što klasične metode nisu mogle.

Međutim, ti modeli su imali ograničenja – bili su spori za treniranje, teško su hvatali dugoročne ovisnosti u tekstu i nisu se dobro skalirali na velike skupove podataka.

Danas su u praksi gotovo potpuno zamijenjeni transformer modelima, koji bolje razumiju kontekst i omogućuju znatno bolje rezultate, pa ćemo se fokusirati upravo na njih.

RNN = obrađuje tekst sekvencijalno LSTM = poboljšani RNN (bolje pamti) problem:

- sporo
- teško pamti duge rečenice
- ne može paralelno procesirati

Transformeri upravo rješavaju te probleme

Klasične metode (BoW, TF-IDF)

I dalje se koriste dosta često

gdje:

- pretraga (search)
- spam filteri
- klasifikacija dokumenata
- manji sustavi bez GPU-a

zašto:

- jednostavno
- brzo
- jeftino
- dovoljno dobro

Uvod u Transformere

Primjer:

Vidio sam jučer lika koji šeće svog pasa, omotao ga remenom oko svog pasa.

Kod Word2Vec-a svaka riječ ima jedan vektor, bez obzira gdje se nalazi. Transformeri rade drugačije — značenje riječi ovisi o kontekstu.

Attention je način da model odredi koje su riječi u rečenici najvažnije za razumijevanje druge riječi.

Vidio sam jučer lika koji šeće svog pasa, omotao ga remenom oko svog pasa.

Koje su riječi najvažnije u prošloj rečenici za razumijevanje riječi pasa?

Model će za prvo “pasa” obratiti pažnju na riječi poput “šeće”, jer upućuju na životinju, a za drugo “pasa” na riječi poput “remenom” i “oko”, koje upućuju na dio tijela.

Značenje riječi ne dolazi iz same riječi, nego iz odnosa s drugim riječima u rečenici.

Ivan je rekao Marku da je on pogriješio.

Tko je on? Ponekad i nama judima nedostaje još lpnteksta tako da u ovom slučaju ni Transformeri ne pomažu.

Kako rade Transformeri?

- model koji koristi attention kao glavni mehanizam
- obrađuje cijelu rečenicu odjednom
- razumije odnose između riječi

ista riječ → različit vektor (ovisno o kontekstu)

Tokenizacija

Tekst se razbija na tokene.

Ali token ne mora uvijek biti jedna riječ, može biti dio riječi ili čak samo jedan znak Uz tokene, dodaje se i positional encoding (informacija o poziciji riječi)

Predtrenirani modeli

U praksi se najčešće koriste predtrenirani modeli

Pipeline u praksi

Model prvo razbije tekst na tokene i svaki token pretvara u broj (ID), jer model ne radi direktno s riječima nego s brojevima. Zatim svaki token dobije početni vektor koji predstavlja njegovo osnovno značenje.

U sljedećem koraku transformer, koristeći attention mehanizam, povezuje sve riječi u rečenici i omogućuje da svaka riječ “vidi” ostale te prilagodi svoje značenje kontekstu. Time nastaju kontekstualne reprezentacije (novi vektori), gdje ista riječ može imati različito značenje ovisno o rečenici.

Na kraju, ovisno o zadatku, model koristi te reprezentacije za donošenje odluke, poput klasifikacije teksta ili generiranja novog teksta.

Demo 1: Sentiment analiza

```
from transformers import pipeline

classifier = pipeline("sentiment-analysis")

print(classifier("I love this course"))
print(classifier("This is not good"))
```

```
No model was supplied, defaulted to distilbert/distilbert-base-uncased-finetuned-sst-2-english and revision 714eb0f.
Using a pipeline without specifying a model name and revision in production is not recommended.
config.json: 100%   629/629 [00:00<00:00, 24.8kB/s]
model.safetensors: 100%   268M/268M [00:05<00:00, 47.5MB/s]
Loading weights: 100%   104/104 [00:00<00:00, 2357.33it/s]
tokenizer_config.json: 100%   48.0/48.0 [00:00<00:00, 3.84kB/s]
vocab.txt:   232k/? [00:00<00:00, 5.79MB/s]
[{'label': 'POSITIVE', 'score': 0.999883770942688}]
[{'label': 'NEGATIVE', 'score': 0.9997923970222473}]
```

```
print(classifier("The movie was not bad at all"))
```

```
[{'label': 'POSITIVE', 'score': 0.9973582625389099}]
```

```
print(classifier("The fries were warm, but too salty"))
```

```
[{'label': 'NEGATIVE', 'score': 0.9804040789604187}]
```

Što je Hugging face?

Hugging Face je open-source Python library i platforma koja omogućuje jednostavno korištenje unaprijed treniranih modela za obradu prirodnog jezika i drugih AI zadataka. Kroz njega možemo u nekoliko linija koda koristiti modele za klasifikaciju teksta, prepoznavanje entiteta, summarization, prijevod, odgovaranje na pitanja i generiranje teksta, bez potrebe za treniranjem modela od nule.

Demo 2: Demo: Named Entity Recognition (NER)

```
from transformers import pipeline

ner = pipeline("ner", aggregation_strategy="simple")

text = "Elon Musk founded SpaceX and Tesla in the United States."

print(ner(text))
```

```
No model was supplied, defaulted to dbmdz/bert-large-cased-finetuned-conll03-english and revision 4c53496.
Using a pipeline without specifying a model name and revision in production is not recommended.
Loading weights: 100%                                391/391 [00:00<00:00, 855.54it/s, Materializing param=classifier.weight]
BertForTokenClassification LOAD REPORT from: dbmdz/bert-large-cased-finetuned-conll03-english
Key                       | Status     |  |
--------------------------+------------+--+-
bert.pooler.dense.weight  | UNEXPECTED |  |
bert.pooler.dense.bias    | UNEXPECTED |  |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
[{'entity_group': 'PER', 'score': np.float32(0.9975515), 'word': 'Elon Musk', 'start': 0, 'end': 9}, {'entity_group': 'ORG', 'score': np.float
```

Demo 3: Zero-shot klasifikacija

```
from transformers import pipeline

classifier = pipeline("zero-shot-classification")

text = "This course teaches machine learning and AI."

labels = ["sports", "technology", "politics"]
```

```
print(classifier(text, labels))
```

```
No model was supplied, defaulted to facebook/bart-large-mnli and revision d7645e1.
Using a pipeline without specifying a model name and revision in production is not recommended.
config.json:  1.15k/? [00:00<00:00, 86.3kB/s]
model.safetensors: 100%  1.63G/1.63G [01:12<00:00, 20.1MB/s]
Loading weights: 100%  515/515 [00:00<00:00, 648.18it/s, Materializing param=model.shared.weight]
tokenizer_config.json: 100%  26.0/26.0 [00:00<00:00, 832B/s]
vocab.json:  899k/? [00:00<00:00, 10.4MB/s]
merges.txt:  456k/? [00:00<00:00, 14.0MB/s]
tokenizer.json:  1.36M/? [00:00<00:00, 29.6MB/s]
{'sequence': 'This course teaches machine learning and AI.', 'labels': ['technology', 'sports', 'politics'], 'scores': [0.9944332838058472, 0.
```

```
Počnite pisati kôd ili generirati uz pomoć umjetne inteligencije.
```

Grupni rad: Zadatak: Istraživanje AI modela (Transformers u praksi)

Podijelite se u grupe. Svaka grupa dobiva jedan AI zadatak iz liste dostupnih pipeline-ova. Vaš cilj je isprobati model, razumjeti što radi i kratko prezentirati rezultate.

available tasks are ['any-to-any', 'audio-classification', 'automatic-speech-recognition', 'depth-estimation', 'document-question-answering', 'feature-extraction', 'fill-mask', 'image-classification', 'image-feature-extraction', 'image-segmentation', 'image-text-to-text', 'keypoint-matching', 'mask-generation', 'ner', 'object-detection', 'sentiment-analysis', 'table-question-answering', 'text-classification', 'text-generation', 'text-to-audio', 'text-to-speech', 'token-classification', 'video-classification', 'zero-shot-audio-classification', 'zero-shot-classification', 'zero-shot-image-classification', 'zero-shot-object-detection']"

odaberite jedan zadatak iz liste

Napravite jednostavan primjer u Pythonu koristeći pipeline() Testirajte model na barem 3 vlastita primjera Pokušajte model “zbuniti”

```
!pip install -U transformers
```

```
Requirement already satisfied: transformers in /usr/local/lib/python3.12/dist-packages (5.0.0)
Collecting transformers
  Downloading transformers-5.5.0-py3-none-any.whl.metadata (32 kB)
Requirement already satisfied: huggingface-hub<2.0,>=1.5.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (1.8.0)
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2.0.2)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (26.0)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from transformers) (6.0.3)
Requirement already satisfied: regex>=2025.10.22 in /usr/local/lib/python3.12/dist-packages (from transformers) (2025.11.3)
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.22.2)
Requirement already satisfied: typer in /usr/local/lib/python3.12/dist-packages (from transformers) (0.24.1)
Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.7.0)
Requirement already satisfied: tqdm>=4.27 in /usr/local/lib/python3.12/dist-packages (from transformers) (4.67.3)
Requirement already satisfied: filelock>=3.10.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.5.0->transformers) (3
Requirement already satisfied: fsspec>=2023.5.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.5.0->transformers) (2
Requirement already satisfied: hf-xet<2.0.0,>=1.4.2 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.5.0->transformers
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.5.0->transformers) (0
Requirement already satisfied: typing-extensions>=4.1.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.5.0->transfor
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-packages (from typer->transformers) (8.3.1)
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->transformers) (1.5.4)
Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->transformers) (13.9.4)
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer->transformers) (0.0.4)
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.5.0->transform
Requirement already satisfied: certifi in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.5.0->transfo
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.5.0->t
Requirement already satisfied: idna in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=1.5.0->transforme
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0->huggingface-hub<2.0
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer->transformers) (4.0.
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer->transformers) (2.
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich>=12.3.0->typer->transfo
Downloading transformers-5.5.0-py3-none-any.whl (10.2 MB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 10.2/10.2 MB 42.7 MB/s eta 0:00:00
Installing collected packages: transformers
  Attempting uninstall: transformers
    Found existing installation: transformers 5.0.0
    Uninstalling transformers-5.0.0:
      Successfully uninstalled transformers-5.0.0
Successfully installed transformers-5.5.0
```

```
from transformers import pipeline

masker = pipeline("fill-mask")

text = "Artificial intelligence is <mask> the world."

result = masker(text)
```

```
for r in result:
    print(r["sequence"], "| score:", round(r["score"], 3))
```

```
No model was supplied, defaulted to distilbert/distilroberta-base and revision fb53ab8.
Using a pipeline without specifying a model name and revision in production is not recommended.
```

Loading weights: 100% 106/106 [00:00<00:00, 1733.91it/s]

```
RobertaForMaskedLM LOAD REPORT from: distilbert/distilroberta-base
Key                         | Status     |  |
----------------------------+------------+--+-
roberta.pooler.dense.bias   | UNEXPECTED |  |
roberta.pooler.dense.weight | UNEXPECTED |  |

Notes:
- UNEXPECTED:   can be ignored when loading from different task/architecture; not ok if you expect identical arch.
Artificial intelligence is changing the world. | score: 0.406
Artificial intelligence is transforming the world. | score: 0.212
Artificial intelligence is sweeping the world. | score: 0.107
Artificial intelligence is saving the world. | score: 0.038
Artificial intelligence is destroying the world. | score: 0.029
```

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